BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 126 · CPU Architecture and the Fetch-Execute Cycle

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a logic error?

Q2. Which description matches an AND gate symbol?
A. Triangle with output circle
B. Flat input; rounded output
C. Curved input; pointed output
D. A decision diamond

Q3. “Iterative testing happens only after the entire finished product is delivered.” Correct this. Explain why.

Q4. A booking program is tested before release. “Final testing means no testing is needed while building it.” Correct this. Why are both needed?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

ALU — Performs arithmetic and logical operations.

Control Unit — Coordinates instruction execution and the CPU's internal operations.

Registers — PC stores next instruction address, MAR stores address accessed, MDR holds transferred data/instruction and ACC holds calculation results.

Fetch–decode–execute — Instruction fetches from memory, decodes, executes and the cycle repeats.

New learning

ALU: Performs arithmetic and logical operations.

Control Unit: Coordinates instruction execution and the CPU's internal operations.

Registers: PC stores next instruction address, MAR stores address accessed, MDR holds transferred data/instruction and ACC holds calculation results.

Fetch–decode–execute: Instruction fetches from memory, decodes, executes and the cycle repeats.

Worked example

Trace how the PC, MAR and MDR participate in fetching an instruction.

1. PC holds the address of the next instruction.
2. The address is copied to MAR.
3. The instruction at that address is transferred to MDR.
4. The instruction is decoded/executed, and PC is updated as appropriate.

We Do · Practise with guidance

Guided 1

Explain how the ALU and accumulator work together for an addition.

Worked method: Control Unit coordinates execution of the instruction.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the final value of the illustrative accumulator variable.

```
acc = 4
acc = acc + 6
print(acc)
```

predict

Question 1

State the purpose of MAR and MDR. [4 marks]

Question 2

What does the Program Counter hold? [1 mark]

Question 3

Explain the role of the Control Unit. [2 marks]

Exit Ticket · Answer before checking

The MDR stores the address of the next instruction.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A mistake in the algorithm that gives the wrong result, even though the code follows the language's rules.

Q2. B — Flat input; rounded output

Q3. Testing and correcting parts during development as the solution is built.

Q4. Final testing checks the finished program against its requirements. Testing during development finds faults earlier. Both are needed.

I Do / We Do pages · Guided practice

Q1. Control Unit coordinates execution of the instruction.

You Do pages · Code First

```
acc = 4
acc = acc + 6
print(acc)
```

Expected result: 10

You Do pages · Independent questions

Q1. MAR: address of memory location accessed; MDR: data/instruction being transferred.

Q2. Address of the next instruction to fetch.

Q3. Coordinates CPU operations and controls execution of instructions.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: